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Sharp linear hypercontractivity for ultraspherical measures

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Abstract

The linear hypercontractive estimate of Ivanisvili, Lindenberger, Müller and Schmuckenschläger on the circle, proved in their work for exponents $6 \le p \le q$, holds for every admissible pair. For $m \ge -1$ and $q \ge p > 0$ with $p + m > 0$,

$$\|1 + rbz\|_{L^q(\nu_m)} \le \|1 + bz\|_{L^p(\nu_m)}$$

for all real b, if and only if $|r| \le \sqrt{(p+m)/(q+m)}$. At $m = -1$ the restriction $p + m > 0$ is Bonami's condition $p > 1$. The same threshold is sharp for the corresponding linear estimate on every sphere.

1 Statement

For $m > -1$ let ν_m be the ultraspherical probability measure on the unit circle,

$$d\nu_m(e^{i\theta}) = c_m \,|\sin\theta|^m \, d\theta, \qquad c_m = \frac{\Gamma(m/2+1)}{2\sqrt{\pi}\,\Gamma(m/2+1/2)},$$

the integral running over $[0, 2\pi]$, and let $\nu_{-1} = \frac{1}{2}(\delta_1 + \delta_{-1})$. The constant c_m is chosen so that ν_m is a probability measure. Indeed,

$$\int_0^{\pi/2} \sin^m\theta \, d\theta = \frac{1}{2} B\Big(\frac{m+1}{2}, \frac{1}{2}\Big) = \frac{\sqrt{\pi}\,\Gamma(m/2+1/2)}{2\,\Gamma(m/2+1)},$$

and four copies of this integral give

$$\int_0^{2\pi} |\sin\theta|^m \, d\theta = 2\sqrt{\pi}\,\frac{\Gamma(m/2+1/2)}{\Gamma(m/2+1)}.$$

The estimate under consideration is

$$\|1 + rbz\|_{L^q(\nu_m)} \le \|1 + bz\|_{L^p(\nu_m)} \qquad \text{for every } b \in \mathbb{R}. \tag{1}$$

Theorem 1. *Let $m \ge -1$ and $q \ge p > 0$ with $p + m > 0$. Then (1) holds for every real b if and only if*

$$|r| \le \sqrt{\frac{p+m}{q+m}}.$$

At $m = -1$, the hypothesis $p + m > 0$ is exactly $p > 1$.

The case $m = -1$ is Bonami's theorem, and the case $m = 0$ is Weissler's theorem on the circle. Theorem 1 removes the restriction $6 \le p \le q$ from Theorem 3.1 of Ivanisvili–Lindenberger–Müller–Schmuckenschläger, and it extends their Theorem 2.1 from the exponents $0 < p \le 2 = q$ on the sphere to every pair $q \ge p > 0$.

Theorem 2. *Let $n \geq 2$, let σ be the uniform probability measure on $\mathbb{S}^{n-1}$, and let $q \geq p > 0$. For every $x \in \mathbb{R}^n$ and every real a,*

$$\|x + ary\|_{L^q(\mathbb{S}^{n-1},\sigma)} \leq \|x + ay\|_{L^p(\mathbb{S}^{n-1},\sigma)}$$

whenever $|r| \leq \sqrt{(p+n-2)/(q+n-2)}$. The constant is sharp.

Throughout, $m = 2\lambda$ with $\lambda > -1/2$, and μ_λ denotes the law of $\operatorname{Re} z$ under ν_m. Explicitly, μ_λ is the probability measure on $[-1, 1]$ with density

$$\frac{\Gamma(\lambda+1)}{\sqrt{\pi}\,\Gamma(\lambda+1/2)}\,(1-t^2)^{\lambda-1/2}.$$

Indeed, $\cos\theta = t$ has two preimages in $[0, 2\pi)$, each with Jacobian $1/\sqrt{1-t^2}$, so the density of the push-forward is

$$2c_{2\lambda}(1-t^2)^{\lambda-1/2},$$

and $2c_{2\lambda}$ is the constant displayed above. For $c \geq 0$ write

$$F_c(t) = 1 + 2ct + c^2$$

and

$$N_\alpha(c) = \left(\int_{-1}^{1} F_c(t)^\alpha \, d\mu_\lambda(t)\right)^{1/\alpha},$$

for every $\alpha > 0$ with $\alpha + \lambda > 0$. Since ν_m is invariant under $z \mapsto -z$,

$$\|1 + bz\|_{L^{2\alpha}(\nu_m)} = N_\alpha(|b|)^{1/2}.$$

2 Necessity

Proposition 3. *If $m > -1$ and $p + m > 0$, $q + m > 0$, and if (1) holds for every small real b, then $|r| \leq \sqrt{(p+m)/(q+m)}$.*

Proof. The identity $\cos^2\theta = 1 - \sin^2\theta$ gives

$$\int (\operatorname{Re} z)^2 \, d\nu_m = 1 - \frac{c_m}{c_{m+2}}.$$

The recurrence $\Gamma(w+1) = w\,\Gamma(w)$ gives

$$\frac{c_m}{c_{m+2}} = \frac{\Gamma(m/2+1)}{\Gamma(m/2+2)} \cdot \frac{\Gamma(m/2+3/2)}{\Gamma(m/2+1/2)} = \frac{m/2+1/2}{m/2+1} = \frac{m+1}{m+2},$$

so the second moment equals $1 - (m+1)/(m+2) = 1/(m+2)$.

For $|b| < 1/2$ one has $1 + 2b\operatorname{Re} z + b^2 \geq (1-|b|)^2 > 1/4$. Taylor expansion with a uniform remainder therefore yields

$$\begin{aligned}\int (1 + 2b\operatorname{Re} z + b^2)^{p/2} \, d\nu_m &= 1 + \frac{p}{2}b^2 + \frac{p(p-2)}{2(m+2)}b^2 + O(b^3) \\ &= 1 + \frac{pb^2}{2} \cdot \frac{p+m}{m+2} + O(b^3).\end{aligned}$$

The odd contributions in b integrate to zero by the symmetry $z \mapsto -z$. Raising to the power $1/p$ gives

$$\|1 + bz\|_{L^p(\nu_m)} = 1 + \frac{b^2}{2} \cdot \frac{p+m}{m+2} + O(b^3).$$

The same expansion at exponent q, with b replaced by rb, gives

$$\|1 + rbz\|_{L^q(\nu_m)} = 1 + \frac{r^2 b^2}{2} \cdot \frac{q+m}{m+2} + O(b^3).$$

Subtract the constant term 1, divide by b^2, and let $b \to 0$. Inequality (1) forces

$$r^2 \frac{q+m}{m+2} \le \frac{p+m}{m+2}.$$

□

The same expansion at $m = -1$, where $\int (\operatorname{Re} z)^2 \, d\nu_{-1} = 1$, gives the necessity of $|r| \le \sqrt{(p-1)/(q-1)}$ for $p, q > 1$.

3 The hypergeometric representation

For $0 \le c < 1$ and every real α,

$$\int F_c^\alpha \, d\mu_\lambda = {}_2F_1(-\alpha, -\alpha - \lambda; \lambda + 1; c^2). \tag{2}$$

The identity $F_c = c^2 F_{1/c}$ gives, for every $c > 0$,

$$N_\alpha(c) = c^2 N_\alpha(1/c). \tag{3}$$

To prove (2), set $z = c^2$ and $y(z) = \int F_{\sqrt{z}}^\alpha \, d\mu_\lambda$. The function

$$Y(z) = {}_2F_1(-\alpha, -\alpha - \lambda; \lambda + 1; z)$$

is the solution of

$$z(1-z)Y'' + \big[\lambda + 1 - (1 - 2\alpha - \lambda)z\big]Y' - \alpha(\alpha + \lambda)Y = 0 \tag{4}$$

which is analytic at the origin and satisfies $Y(0) = 1$. The constant term of y is 1. The coefficient of z is computed from

$$F_c^\alpha = 1 + \alpha(b^2 + 2bt) + \alpha(\alpha - 1)2b^2 t^2 + O(b^3),$$

with $b = \sqrt{z}$. The remainder $O(b^3)$ is $O(z^{3/2})$, hence $o(z)$. Together with $\int t \, d\mu_\lambda = 0$ and

$$\int t^2 \, d\mu_\lambda = \frac{1}{2(\lambda + 1)}.$$

(The second moment is the case $m = 2\lambda$ of the computation in Proposition 3.) Integrating and using $\int t \, d\mu_\lambda = 0$ produces

$$y(z) = 1 + \alpha z + \frac{\alpha(\alpha - 1)}{\lambda + 1} \, z + o(z) = 1 + \frac{\alpha(\alpha + \lambda)}{\lambda + 1} \, z + o(z),$$

so

$$y'(0) = \frac{\alpha(\alpha+\lambda)}{\lambda+1}.$$

The derivative of ${}_2F_1(a,b;c_*;z)$ at the origin equals ab/c_*. With $a=-\alpha$, $b=-\alpha-\lambda$ and $c_*=\lambda+1$, this is the same value. The map $c \mapsto \int F_c^\alpha \, d\mu_\lambda$ is analytic for $|c|<1$, because $F_c \geq (1-|c|)^2 > 0$, and it is even by the substitution $t \mapsto -t$. An even analytic function of c is analytic in c^2, so y is analytic on $[0,1)$. The hypergeometric series is the unique analytic solution of (4) with value 1 at the origin: the constant term of the differential equation forces the coefficient of z, and each higher coefficient is then determined by the recurrence. Thus it is enough to show that y satisfies (4) on $(0,1)$.

Lemma 4. *Let $c \in \mathbb{R}$ and let $F = F_c$. On $(-1,1)$,*

$$\frac{d}{dt}\Big(F^{\alpha-1}(1-t^2)^{\lambda+1/2}\Big) = -Q\,F^{\alpha-2}(1-t^2)^{\lambda-1/2},$$

where

$$Q = (2\lambda+1)(1+c^2)\,t + 2c(\alpha+2\lambda)\,t^2 + 2c(1-\alpha).$$

Proof. Differentiate the product. With $F' = 2c$,

$$\begin{aligned}\frac{d}{dt}\Big(F^{\alpha-1}(1-t^2)^{\lambda+1/2}\Big) &= 2c(\alpha-1)F^{\alpha-2}(1-t^2)^{\lambda+1/2} - t(2\lambda+1)F^{\alpha-1}(1-t^2)^{\lambda-1/2} \\ &= F^{\alpha-2}(1-t^2)^{\lambda-1/2}\Big(2c(\alpha-1)(1-t^2) - t(2\lambda+1)F\Big).\end{aligned}$$

The expression in parentheses equals $-Q$, because

$$\begin{aligned}t(2\lambda+1)F - 2c(\alpha-1)(1-t^2) &= (2\lambda+1)(1+c^2)t + 2c(2\lambda+1)t^2 + 2c(\alpha-1)t^2 - 2c(\alpha-1) \\ &= (2\lambda+1)(1+c^2)t + 2c(\alpha+2\lambda)t^2 + 2c(1-\alpha).\end{aligned}$$

□

For $c \in (0,1)$ one has $F \geq (1-c)^2 > 0$, so $F^{\alpha-1}$ is bounded on $[-1,1]$. The boundary factor $(1-t^2)^{\lambda+1/2}$ vanishes at $t=\pm1$ because $\lambda > -1/2$. The derivative on the left of Lemma 4 is integrable on $(-1,1)$, since near each endpoint it is comparable to $(1-t^2)^{\lambda-1/2}$ and $\lambda - 1/2 > -1$. Consequently

$$\int Q\,F^{\alpha-2}\,d\mu_\lambda = 0.$$

Now let $I(c) = \int F_c^\alpha \, d\mu_\lambda$, so that $y(z) = I(\sqrt{z})$. Then

$$y_z = \frac{I_c}{2c}, \qquad y_{zz} = \frac{I_{cc}}{4c^2} - \frac{I_c}{4c^3}.$$

Substituting $z = c^2$ into the differential expression (4) and clearing the denominator $4c^3$ produces, under the integral sign, the integrand identity

$$\begin{aligned}&z(1-z)\partial_{zz}(F^\alpha) + \big[\lambda+1-(1-2\alpha-\lambda)z\big]\partial_z(F^\alpha) - \alpha(\alpha+\lambda)F^\alpha \\ &\quad = -\frac{\alpha(c^2-1)}{2c}\,Q\,F^{\alpha-2}.\end{aligned} \tag{5}$$

(The two sides are rational functions of c and t, once $F^{\alpha-2}$ is divided out, and the resulting polynomial identity is the expansion of Lemma 4 against the chain rule; it is recorded in the appendix.) Integrating (5) and using $\int QF^{\alpha-2}\,d\mu_\lambda = 0$ shows that y satisfies (4). This proves (2).

For $\alpha > 0$ and $z \in (0,1)$ define

$$\tau_\alpha(z) = \frac{1}{\alpha}\,\partial_z \log y(\alpha, z) = \frac{\partial_z y(\alpha, z)}{\alpha\, y(\alpha, z)},$$

where $y(\alpha, z)$ denotes the right-hand side of (2). Then τ_α extends continuously to $z = 0$ by

$$\tau_\alpha(0) = \frac{\alpha + \lambda}{\lambda + 1},$$

and the substitution $\partial_z \log y = \alpha\tau_\alpha$ turns (4) into the Riccati equation

$$z(1-z)\tau' = (\alpha + \lambda) - \big[\lambda + 1 - (1 - 2\alpha - \lambda)z\big]\tau - z(1-z)\,\alpha\,\tau^2. \tag{6}$$

In particular,

$$\frac{1}{\alpha}\log y(\alpha, z) = \int_0^z \tau_\alpha(t)\,dt. \tag{7}$$

Lemma 5. *If $\alpha > 0$ and $\alpha + \lambda > 0$, then $\tau_\alpha(z) > 0$ for every $z \in [0, 1)$.*

Proof. The value at the origin is positive. If $z \in (0,1)$ were a first zero, (6) at $\tau = 0$ would give $z(1-z)\tau' = \alpha + \lambda > 0$, so $\tau' > 0$, which is incompatible with reaching zero from above. □

4 An upper bound for the logarithmic derivative

Lemma 6. *For every $\lambda > -1/2$, every $\alpha > 0$ and every $z \in (0,1)$,*

$$\tau_\alpha(z) < \frac{1}{\sqrt{z}\,(1 + \sqrt{z})}.$$

In particular $\tau_\alpha(z) < 1/(2z)$.

Proof. Write $\sigma(z) = 1/(\sqrt{z}\,(1+\sqrt{z}))$ and $s = \sqrt{z}$, so that $\sigma(z) = 1/(s(s+1))$. As $z \downarrow 0$ one has $\sigma(z) \to \infty$, while $\tau_\alpha(z)$ tends to the finite value $(\alpha + \lambda)/(\lambda + 1)$. Thus $\tau_\alpha < \sigma$ on a right neighbourhood of the origin.

Set $t = 1/(s(s+1))$ and substitute $\tau = t$, $z = s^2$ into the right-hand side of (6). The quadratic term contributes

$$z(1-z)\,\alpha\,\tau^2 \cdot s(s+1) = \alpha\, s^2(1-s^2) \cdot \frac{1}{s(s+1)} = \alpha\, s(1-s),$$

where $1 - s^2 = (1-s)(1+s)$ and $(1+s)/(s+1) = 1$. Multiplying the whole right-hand side by $s(s+1)$ therefore gives

$$\begin{aligned}(\alpha + \lambda)s(s+1) - \big[\lambda + 1 - (1 - 2\alpha - \lambda)s^2\big] - \alpha s(1-s)\\ = s^2 + \lambda s - (\lambda + 1) = (s-1)(s + \lambda + 1).\end{aligned}$$

(The coefficient of s^2 is $(\alpha+\lambda) + (1 - 2\alpha - \lambda) + \alpha = 1$, and the coefficient of s is $(\alpha+\lambda) - \alpha = \lambda$.) Therefore, at every point where $\tau_\alpha = \sigma$,

$$\tau_\alpha' = \frac{(s-1)(s+\lambda+1)}{s(s+1)\cdot s^2(1-s^2)} = -\frac{s + \lambda + 1}{s^3(s+1)^2},$$

where $1-s^2=(1-s)(1+s)$ and $s-1=-(1-s)$ were used. On the other hand,

$$\frac{d}{ds}\frac{1}{s(s+1)}=-\frac{1+2s}{s^2(s+1)^2},\qquad \frac{ds}{dz}=\frac{1}{2s},$$

so

$$\sigma'(z)=-\frac{1+2s}{2s^3(s+1)^2}.$$

Subtracting these evaluations yields

$$\tau_\alpha'-\sigma'=\frac{-2(s+\lambda+1)+(1+2s)}{2s^3(s+1)^2}=-\frac{2\lambda+1}{2s^3(s+1)^2}<0,$$

because $\lambda>-1/2$. At a first zero of $\sigma-\tau_\alpha$ in $(0,1)$ one would have $\tau_\alpha'\geq\sigma'$. This contradiction keeps $\tau_\alpha<\sigma$ on the whole interval.

Finally, $z+\sqrt{z}\geq 2z$ for $z\in(0,1]$, so $\sigma(z)\leq 1/(2z)$, and the inequality is strict for $z\in(0,1)$. □

Lemma 7. *For every $\alpha>0$ with $\alpha+\lambda>0$, the map $c\mapsto N_\alpha(c)$ is continuous and strictly increasing on $[0,\infty)$.*

Proof. For $c\in(0,1)$, (2) and the chain rule give

$$\frac{N_\alpha'(c)}{N_\alpha(c)}=2c\,\tau_\alpha(c^2).$$

Lemma 5 makes the right-hand side positive. For $c>1$, (3) gives

$$\frac{d}{dc}\log N_\alpha(c)=\frac{2}{c}-\frac{1}{c^2}\cdot\frac{N_\alpha'(1/c)}{N_\alpha(1/c)}=\frac{2}{c}-\frac{2}{c^3}\,\tau_\alpha(c^{-2}).$$

Lemma 6 yields $\tau_\alpha(c^{-2})<c^2/2$, so the derivative is positive. Continuity on $[0,\infty)$ holds because $0\leq F_c\leq(1+c)^2$ and $\alpha>0$. □

Lemma 7 removes the restriction to the critical multiplier. The law of z is symmetric, so b may be replaced by $|b|$. For $0\leq r\leq r_*$ and $c\geq 0$,

$$N_{q/2}(rc)\leq N_{q/2}(r_*c).$$

Thus (1) holds for every r with $|r|\leq\sqrt{(p+m)/(q+m)}$ as soon as it holds at the positive endpoint.

5 Scales at most one

Set $\gamma=p/2$ and $\delta=q/2$. The hypotheses $q\geq p>0$ and $p+m>0$ become $0<\gamma\leq\delta$ and $\gamma+\lambda>0$. Set

$$\kappa=\frac{\gamma+\lambda}{\delta+\lambda}\in(0,1].$$

Inequality (1) at the critical multiplier is equivalent to

$$N_\gamma(c)\geq N_\delta(c\sqrt{\kappa})\qquad\text{for every } c\geq 0. \tag{8}$$

Lemma 8. *For every $z\in(0,1)$,*

$$\tau_\gamma(z)>\kappa\,\tau_\delta(\kappa z)$$

whenever $\delta>\gamma$, and therefore $N_\gamma(\sqrt{z})\geq N_\delta(\sqrt{\kappa z})$. In particular, (8) holds for every $c\in[0,1]$.

Proof. The case $\gamma = \delta$ is the identity $\kappa = 1$, so assume $\delta > \gamma$. Write $\psi(z) = \kappa\,\tau_\delta(\kappa z)$. Then $\tau_\gamma(0) = \psi(0) = (\gamma+\lambda)/(\lambda+1)$. Differentiate (6). The derivative of the left side is $(1-2z)\tau' + z(1-z)\tau''$, which equals $\tau'(0)$ at $z = 0$. At $z = 0$ and $\tau = t_0 = (\alpha+\lambda)/(\lambda+1)$ both sides of (6) vanish. Differentiating the right side, the product rule contributes $(1-2\alpha-\lambda)\tau$ from the coefficient of τ and $-\alpha\tau^2$ from $z(1-z)\alpha\tau^2$, together with the terms that still contain τ'. At the origin this equals

$$(1-2\alpha-\lambda)t_0 - (\lambda+1)\tau'(0) - \alpha t_0^2.$$

Solving for the derivative gives

$$\tau'_\alpha(0) = \frac{t_0}{\lambda+2}\big(1-2\alpha-\lambda-\alpha t_0\big). \tag{9}$$

The chain rule yields $\psi'(0) = \kappa^2\tau'_\delta(0)$. Since $\kappa t_\delta = t_\gamma$, with $t_\alpha = (\alpha+\lambda)/(\lambda+1)$,

$$\tau'_\gamma(0) - \psi'(0) = \frac{t_\gamma}{\lambda+2}\Big((1-2\gamma-\lambda-\gamma t_\gamma) - \kappa(1-2\delta-\lambda-\delta t_\delta)\Big).$$

The expression in parentheses equals

$$\frac{\delta-\gamma}{(\delta+\lambda)(\lambda+1)}\big(\gamma\delta + \lambda(\gamma+\delta) + 2\lambda^2 + 2\lambda + 1\big).$$

(The verification is the expansion in the appendix.) The quadratic form factors as

$$\gamma\delta + \lambda(\gamma+\delta) + 2\lambda^2 + 2\lambda + 1 = (\gamma+\lambda)(\delta+\lambda) + (\lambda+1)^2 > 0.$$

Hence $\tau'_\gamma(0) - \psi'(0) > 0$. Combined with $\tau_\gamma(0) = \psi(0)$, this forces $\tau_\gamma > \psi$ on a right neighbourhood of the origin.

Suppose there exists a point of $(0,1)$ at which $\tau_\gamma = \psi$. Let z_* be the infimum of such points. Then $z_* > 0$, $\tau_\gamma(z_*) = \psi(z_*)$, and $\tau_\gamma - \psi \geq 0$ on $[0, z_*]$, so the derivative of the difference at z_* is at most zero. It remains to rule that out.

Let $z \in (0,1)$ satisfy $\tau_\gamma(z) = \psi(z) = t$. The equation (6) at exponent γ reads

$$z(1-z)\tau'_\gamma = N_\gamma(t),$$

where

$$N_\gamma(t) = (\gamma+\lambda) - \big[\lambda+1-(1-2\gamma-\lambda)z\big]t - z(1-z)\gamma t^2.$$

For the comparison function, set $u = \kappa z$ and $\tau_\delta(u) = \psi(z)/\kappa$. Then $\psi'(z) = \kappa^2\, d\tau_\delta/du$. Substitute into (6) at exponent δ, multiply through by κ^2, and use $\kappa^2(\delta+\lambda) = \kappa(\gamma+\lambda)$. Dividing by $\kappa > 0$ produces

$$z(1-\kappa z)\psi' = N_\psi(t), \tag{10}$$

where

$$N_\psi(t) = (\gamma+\lambda) - \big[\lambda+1-(1-2\delta-\lambda)\kappa z\big]t - z(1-\kappa z)\delta t^2.$$

Therefore

$$\tau'_\gamma(z) - \psi'(z) = \frac{N_\gamma(t)\,(1-\kappa z) - N_\psi(t)\,(1-z)}{z(1-z)(1-\kappa z)}.$$

The denominator is positive. The numerator equals

$$\frac{z(\delta-\gamma)}{\delta+\lambda}\,Q(t),$$

where

$$Q(t) = A_Q t^2 - 2z(\gamma+\lambda)\,t + (\gamma+\lambda), \qquad A_Q = (1-z)\big(\delta+\lambda - z(\gamma+\lambda)\big).$$

This polynomial identity is checked coefficient-wise in the appendix. The prefactor $z(\delta-\gamma)/(\delta+\lambda)$ is positive, so the sign of $\tau_\gamma' - \psi'$ is the sign of $Q(t)$.

Since $\delta \geq \gamma$ and $z \in (0,1)$,

$$A_Q - (1-z)^2(\gamma+\lambda) = (1-z)(\delta-\gamma) \geq 0,$$

and hence

$$Q(t) \geq (\gamma+\lambda)\big((1-z)^2t^2 - 2zt + 1\big).$$

For $z \in (0, 1/2]$ the discriminant of the quadratic in t is $4(2z-1) \leq 0$, so the quadratic is positive for every finite t except at the single point $z = 1/2$, $t = 2$, where it vanishes. For $z \in (1/2, 1)$ the roots are

$$\frac{1}{z \pm \sqrt{2z-1}},$$

because

$$\big(z - \sqrt{2z-1}\big)\big(z + \sqrt{2z-1}\big) = (1-z)^2.$$

The quadratic is then nonnegative for $0 \leq t \leq 1/(z+\sqrt{2z-1})$. Lemma 6 supplies the stricter bound

$$t = \tau_\gamma(z) < \frac{1}{z+\sqrt{z}} \leq \frac{1}{z+\sqrt{2z-1}}$$

on $[1/2, 1)$, the comparison of the denominators being $\sqrt{z} \geq \sqrt{2z-1}$. At $z = 1/2$ the same bound gives $\tau_\gamma(1/2) < 1/(1/2+\sqrt{1/2}) < 2$. Thus $Q(\tau_\gamma(z)) > 0$, and $\tau_\gamma' > \psi'$ at every contact point in $(0,1)$.

The graphs cannot meet in $(0,1)$. Therefore $\tau_\gamma(z) > \psi(z)$ for every $z \in (0,1)$. Integrating and applying (7) gives

$$\frac{1}{\gamma}\log y(\gamma, z) \geq \frac{1}{\delta}\log y(\delta, \kappa z),$$

which is $N_\gamma(\sqrt{z}) \geq N_\delta(\sqrt{\kappa z})$. The endpoint $c = 1$ follows by continuity of N_α. □

6 Scales larger than one

Lemma 9. *For each fixed $\alpha > 0$ with $\alpha + \lambda > 0$, the map $a \mapsto N_\alpha(a)/a$ is strictly decreasing on $(0,1]$.*

Proof. For $a \in (0,1)$,

$$\frac{d}{da}\log\frac{N_\alpha(a)}{a} = 2a\,\tau_\alpha(a^2) - \frac{1}{a}.$$

Lemma 6 yields $2a\,\tau_\alpha(a^2) < 2a\cdot(1/(2a^2)) = 1/a$, so the derivative is negative. If $0 < a < 1$, the same derivative on $(a,1)$ gives $N_\alpha(a)/a > N_\alpha(1)/1$. Thus the map is strictly decreasing on $(0,1]$. □

In particular, $0 < b < a \leq 1$ implies $N_\alpha(b)/b > N_\alpha(a)/a$.

Proposition 10. *Inequality* (8) *holds for every $c > 1$.*

Proof. Write $\rho(c) = N_\gamma(c)/N_\delta(c\sqrt{\kappa})$. Lemma 8 at scale $1/c \in (0,1)$ gives

$$N_\gamma(1/c) \geq N_\delta\big((1/c)\sqrt{\kappa}\big).$$

Suppose first that $c\sqrt{\kappa} > 1$, and set $a = 1/(c\sqrt{\kappa}) \in (0,1)$. Then $(1/c)\sqrt{\kappa} = \kappa a$. The scaling relation (3) gives

$$\rho(c) = \frac{N_\gamma(1/c)}{\kappa\, N_\delta(a)} = \frac{N_\gamma(1/c)}{N_\delta(\kappa a)} \cdot \frac{N_\delta(\kappa a)}{\kappa\, N_\delta(a)}.$$

The first factor is at least 1 by Lemma 8. If $\kappa < 1$, then $0 < \kappa a < a < 1$, so Lemma 9 makes the second factor strictly larger than 1. If $\kappa = 1$, both factors equal 1. Thus $\rho(c) \geq 1$.

Suppose instead that $c > 1$ and $c\sqrt{\kappa} \leq 1$. Then necessarily $\kappa < 1$. Set $x = c\sqrt{\kappa} \in (\sqrt{\kappa}, 1]$ and $y = \kappa/x$. The inequality $x > \sqrt{\kappa}$ forces $y < x$, and both points lie in $(0,1]$. Also $y = (1/c)\sqrt{\kappa}$ and $c^2 = x^2/\kappa$. Hence

$$\rho(c) = \frac{c^2 N_\gamma(1/c)}{N_\delta(x)} \geq \frac{x^2}{\kappa} \cdot \frac{N_\delta(y)}{N_\delta(x)},$$

where the inequality is Lemma 8. The relation $\kappa = xy$ rewrites the desired bound $x^2 N_\delta(y) \geq \kappa N_\delta(x)$ as

$$\frac{N_\delta(y)}{y} \geq \frac{N_\delta(x)}{x},$$

which is Lemma 9. Therefore $\rho(c) \geq 1$. □

Lemmas 8 and 10 are (8) for every $c \geq 0$. By the reduction of Section 7 below, this is Theorem 1 for $m > -1$.

7 The endpoint and the sphere

For $m > -1$, the identity $\|1 + bz\|_{L^{2\alpha}(\nu_m)} = N_\alpha(|b|)^{1/2}$ converts (8) into (1) at

$$r = \sqrt{\kappa} = \sqrt{\frac{p+m}{q+m}}.$$

Lemma 7 extends the inequality to every smaller multiplier, and Proposition 3 gives the matching necessity.

If $m = -1$ and $p > 1$, then $p + m' > 0$ for every $m' \in (-1, 0)$. The case $m > -1$ of the theorem, already proved, gives (1) for $\nu_{m'}$ at the multiplier $\sqrt{(p+m')/(q+m')}$. As $m' \downarrow -1$,

$$\int (\operatorname{Re} z)^2 \, d\nu_{m'} = \frac{1}{m'+2} \to 1.$$

Every weak-star limit point ν satisfies $\int (\operatorname{Re} z)^2 \, d\nu = 1$. Then $\int \big(1 - (\operatorname{Re} z)^2\big) \, d\nu = 0$. The integrand is continuous and nonnegative, so ν is supported on $\{\pm 1\}$. Each $\nu_{m'}$ is invariant under $z \mapsto -z$, so the same is true of ν, and therefore $\nu = \nu_{-1}$. The circle is compact, so $\nu_{m'}$ converges weak-star to ν_{-1}. Fix $b \in \mathbb{R}$ and set $r(m') = \sqrt{(p+m')/(q+m')}$. Then $r(m')$ tends to $r_* = \sqrt{(p-1)/(q-1)}$, and $z \mapsto |1 + r(m')bz|^q$ converges uniformly to $z \mapsto |1 + r_* bz|^q$. The function $z \mapsto |1 + bz|^p$ is continuous. Weak-star convergence preserves both integrals, so the norms pass to the limit and (1) holds for ν_{-1} at the multiplier r_*. The expansion recorded after Proposition 3 gives the matching necessity.

Proof of Theorem 2. If $x = 0$, then $|ay|$ is the constant $|a|$ for every $y \in \mathbb{S}^{n-1}$, and the asserted bound reduces to $|r| \le 1$. This holds because $q \ge p$. Both sides are homogeneous of degree one in (x, a), so one may assume $|x| = 1$. For $y \in \mathbb{S}^{n-1}$,

$$|x + ay|^2 = 1 + 2a\langle x, y\rangle + a^2.$$

The set $\{y \in \mathbb{S}^{n-1} : \langle x, y\rangle \in dt\}$ has area proportional to $(1 - t^2)^{(n-3)/2}\, dt$. With $\lambda = (n-2)/2$, the exponent $(n-3)/2$ equals $\lambda - 1/2$, so the law of $\langle x, y\rangle$ is μ_λ, and the corresponding circle weight is $m = n - 2$. Symmetry in t gives

$$\|x + ay\|_{L^p(\sigma)} = N_{p/2}(|a|)^{1/2},$$

and the same identity with a replaced by ar at the exponent q. Theorem 1 at weight $m = n - 2$ is therefore the spherical estimate. Sharpness is Proposition 3 at this weight. □

Appendix: the polynomial identities

The integrand form of the hypergeometric operator

Let $z = c^2$, $F = 1 + 2ct + c^2$ and $\partial_c F = 2(c + t)$. Write $L[F^\alpha]$ for the left-hand side of (5). The conversions $\partial_z = (1/(2c))\partial_c$ and

$$\partial_{zz} = \frac{1}{4c^2}\partial_{cc} - \frac{1}{4c^3}\partial_c$$

give, after multiplication by $4c^3F^{2-\alpha}$,

$$\begin{aligned}\partial_c(F^\alpha)\, F^{2-\alpha} &= 2\alpha(c + t)F,\\ \partial_{cc}(F^\alpha)\, F^{2-\alpha} &= 4\alpha(\alpha - 1)(c + t)^2 + 2\alpha F.\end{aligned}$$

The multiplied operator is the quadratic polynomial

$$\begin{aligned}E = (1 - c^2)\big(c^3\,\partial_{cc}^{\mathrm{cl}} - c^2\,\partial_c^{\mathrm{cl}}\big)\\ + 2c^2\big[\lambda + 1 - (1 - 2\alpha - \lambda)c^2\big]\,\partial_c^{\mathrm{cl}} - 4\alpha(\alpha + \lambda)c^3F^2,\end{aligned} \tag{11}$$

where the cleared derivatives are the two displays above. Their coefficients in t are

$$\begin{aligned}(c + t)F &= 2c\,t^2 + (1 + 3c^2)t + c(1 + c^2),\\ \partial_c^{\mathrm{cl}} &= 2\alpha\big(2c\,t^2 + (1 + 3c^2)t + c(1 + c^2)\big),\\ \partial_{cc}^{\mathrm{cl}} &= 4\alpha(\alpha - 1)\,t^2 + 4\alpha c(2\alpha - 1)\,t + 2\alpha\big(1 + (2\alpha - 1)c^2\big).\end{aligned}$$

The linear coefficient of $\partial_{cc}^{\mathrm{cl}}$ is $8\alpha(\alpha-1)c + 4\alpha c = 4\alpha c(2\alpha - 1)$, and its constant term is $4\alpha(\alpha-1)c^2 + 2\alpha(1 + c^2) = 2\alpha\big(1 + (2\alpha - 1)c^2\big)$. The square $F^2 = (1 + c^2 + 2ct)^2$ has coefficients $4c^2$, $4c(1 + c^2)$ and $(1 + c^2)^2$.

Coefficient of t^2. The combination $c^3\partial_{cc}^{\mathrm{cl}} - c^2\partial_c^{\mathrm{cl}}$ has t^2-coefficient $4\alpha c^3(\alpha-1) - 4\alpha c^3 = 4\alpha c^3(\alpha - 2)$. Multiplication by $1 - c^2$ contributes $4\alpha c^3(\alpha - 2)(1 - c^2)$. The middle summand of (11) contributes $8\alpha c^3\big[\lambda + 1 - (1 - 2\alpha - \lambda)c^2\big]$, and the last summand contributes $-16\alpha(\alpha + \lambda)c^5$. Dividing the sum by $4\alpha c^3$ produces

$$(\alpha - 2)(1 - c^2) + 2\big[\lambda + 1 - (1 - 2\alpha - \lambda)c^2\big] - 4(\alpha + \lambda)c^2.$$

The terms independent of c^2 are $\alpha - 2 + 2\lambda + 2 = \alpha + 2\lambda$. The coefficient of c^2 is

$$-(\alpha - 2) - 2(1 - 2\alpha - \lambda) - 4(\alpha + \lambda) = -\alpha + 2 - 2 + 4\alpha + 2\lambda - 4\alpha - 4\lambda = -(\alpha + 2\lambda).$$

The quotient equals $(\alpha + 2\lambda)(1 - c^2)$, so the coefficient of t^2 in E is $-4\alpha c^3(\alpha + 2\lambda)(c^2 - 1)$.

Coefficient of t. The three contributions are

$$\begin{aligned}&(1 - c^2)\big(4\alpha(2\alpha - 1)c^4 - 2\alpha c^2(1 + 3c^2)\big),\\ &4\alpha c^2(1 + 3c^2)\big[\lambda + 1 - (1 - 2\alpha - \lambda)c^2\big],\\ &-16\alpha(\alpha + \lambda)c^4(1 + c^2).\end{aligned}$$

Dividing by $2\alpha c^2$ produces

$$(1 - c^2)\big((4\alpha - 5)c^2 - 1\big) + 2(1 + 3c^2)\big[\lambda + 1 - (1 - 2\alpha - \lambda)c^2\big] - 8(\alpha + \lambda)c^2(1 + c^2).$$

The first summand equals $-(4\alpha - 5)c^4 + 4(\alpha - 1)c^2 - 1$. The second equals $2(\lambda + 1) + 6(\lambda + 1)c^2 - 2(1 - 2\alpha - \lambda)c^2 - 6(1 - 2\alpha - \lambda)c^4$. The third equals $-8(\alpha + \lambda)c^2 - 8(\alpha + \lambda)c^4$. The coefficient of c^4 in the sum is

$$\begin{aligned}&-(4\alpha - 5) - 6(1 - 2\alpha - \lambda) - 8(\alpha + \lambda)\\ &\quad = -4\alpha + 5 - 6 + 12\alpha + 6\lambda - 8\alpha - 8\lambda = -(2\lambda + 1).\end{aligned}$$

The coefficient of c^2 is

$$\begin{aligned}&4(\alpha - 1) + 6(\lambda + 1) - 2(1 - 2\alpha - \lambda) - 8(\alpha + \lambda)\\ &\quad = 4\alpha - 4 + 6\lambda + 6 - 2 + 4\alpha + 2\lambda - 8\alpha - 8\lambda = 0.\end{aligned}$$

The remaining term is $-1 + 2(\lambda + 1) = 2\lambda + 1$. The quotient is therefore $(2\lambda + 1)(1 - c^4) = (2\lambda + 1)(1 - c^2)(1 + c^2)$, and the coefficient of t in E is $-2\alpha c^2(c^2 - 1)(c^2 + 1)(2\lambda + 1)$.

Constant term. The three contributions are $4\alpha(\alpha - 1)c^5(1 - c^2)$, $4\alpha c^3(1 + c^2)\big[\lambda + 1 - (1 - 2\alpha - \lambda)c^2\big]$ and $-4\alpha(\alpha + \lambda)c^3(1 + c^2)^2$. Dividing by $4\alpha c^3$ produces

$$(\alpha - 1)c^2(1 - c^2) + (1 + c^2)\big[\lambda + 1 - (1 - 2\alpha - \lambda)c^2\big] - (\alpha + \lambda)(1 + c^2)^2.$$

The coefficient of c^4 is $-(\alpha - 1) - (1 - 2\alpha - \lambda) - (\alpha + \lambda) = 0$. The coefficient of c^2 is

$$(\alpha - 1) + (\lambda + 1) - (1 - 2\alpha - \lambda) - 2(\alpha + \lambda) = \alpha - 1 + \lambda + 1 - 1 + 2\alpha + \lambda - 2\alpha - 2\lambda = \alpha - 1,$$

and the remaining term is $(\lambda + 1) - (\alpha + \lambda) = 1 - \alpha$. The quotient equals $(\alpha - 1)(c^2 - 1)$, so the constant term of E is $4\alpha c^3(\alpha - 1)(c^2 - 1)$.

These are the coefficients of $-2\alpha c^2(c^2 - 1)Q$. Both sides are quadratic polynomials in t, so $E = -2\alpha c^2(c^2 - 1)Q$. The divisions by α and by powers of c compare coefficients of polynomials and do not restrict the identity: both sides vanish to at least those orders. Dividing by $4c^3F^{2-\alpha}$ gives (5).

The initial slope

Start from (9), which is the differentiation of (6) at $z = 0$. With $t_\alpha = (\alpha + \lambda)/(\lambda + 1)$,

$$\begin{aligned}(1 - 2\alpha - \lambda - \alpha t_\alpha)(\lambda + 1) &= (1 - 2\alpha - \lambda)(\lambda + 1) - \alpha(\alpha + \lambda)\\ &= 1 - \lambda^2 - 2\alpha - 2\alpha\lambda - \alpha^2 - \alpha\lambda = M(\alpha),\end{aligned}$$

where $M(\alpha) = (1-\lambda^2) - \alpha^2 - \alpha(3\lambda+2)$. Thus $1 - 2\alpha - \lambda - \alpha t_\alpha = M(\alpha)/(\lambda+1)$. With $\kappa t_\delta = t_\gamma$,

$$(1 - 2\gamma - \lambda - \gamma t_\gamma) - \kappa(1 - 2\delta - \lambda - \delta t_\delta) = \frac{M(\gamma)}{\lambda+1} - \frac{\gamma+\lambda}{\delta+\lambda} \cdot \frac{M(\delta)}{\lambda+1}.$$

The numerator of the common-denominator form is

$$(\delta+\lambda)M(\gamma) - (\gamma+\lambda)M(\delta).$$

The summand $(1-\lambda^2)$ contributes $(\delta-\gamma)(1-\lambda^2)$. The quadratic summand contributes

$$\begin{aligned} -(\delta+\lambda)\gamma^2 + (\gamma+\lambda)\delta^2 &= \gamma\delta(\delta-\gamma) + \lambda(\delta^2-\gamma^2) \\ &= (\delta-\gamma)\big(\gamma\delta + \lambda(\gamma+\delta)\big). \end{aligned}$$

The remaining summand contributes

$$(3\lambda+2)\big(-(\delta+\lambda)\gamma + (\gamma+\lambda)\delta\big) = (3\lambda+2)\big(-\lambda\gamma + \lambda\delta\big) = \lambda(3\lambda+2)(\delta-\gamma).$$

These are the three contributions

$$(1-\lambda^2)(\delta-\gamma), \qquad (\delta-\gamma)\big(\gamma\delta + \lambda(\gamma+\delta)\big), \qquad \lambda(3\lambda+2)(\delta-\gamma).$$

Their sum is

$$(\delta-\gamma)\big(1 + 2\lambda + 2\lambda^2 + \gamma\delta + \lambda(\gamma+\delta)\big) = (\delta-\gamma)\big((\gamma+\lambda)(\delta+\lambda) + (\lambda+1)^2\big).$$

Dividing by $(\delta+\lambda)(\lambda+1)$ and multiplying by $t_\gamma/(\lambda+2)$ gives

$$\tau'_\gamma(0) - \psi'(0) = \frac{(\delta-\gamma)(\gamma+\lambda)\big((\gamma+\lambda)(\delta+\lambda) + (\lambda+1)^2\big)}{(\delta+\lambda)(\lambda+1)^2(\lambda+2)}.$$

Coefficients of the contact polynomial

Write $P(t) = N_\gamma(t)(1-\kappa z) - N_\psi(t)(1-z)$ and

$$S = \frac{z(\delta-\gamma)}{\delta+\lambda}.$$

The constant terms contribute

$$(\gamma+\lambda)\big((1-\kappa z) - (1-z)\big) = (\gamma+\lambda)z(1-\kappa) = S(\gamma+\lambda),$$

since $1-\kappa = (\delta-\gamma)/(\delta+\lambda)$.

The coefficients of t^2 contribute

$$\begin{aligned} &- z(1-z)\gamma(1-\kappa z) + z(1-\kappa z)\delta(1-z) \\ &\quad = z(1-z)(1-\kappa z)(\delta-\gamma). \end{aligned}$$

The relation $1-\kappa z = (\delta+\lambda - z(\gamma+\lambda))/(\delta+\lambda)$ rewrites the right-hand side as SA_Q.

The coefficient of t in N_γ is $-\big[\lambda+1-(1-2\gamma-\lambda)z\big]$, and the coefficient of t in N_ψ is $-\big[\lambda+1-(1-2\delta-\lambda)\kappa z\big]$. Write $A = \lambda+1$, $G = 1-2\gamma-\lambda$ and $D = 1-2\delta-\lambda$. The coefficient of t in P is

$$\begin{aligned} &- (A - Gz)(1-\kappa z) + (A - D\kappa z)(1-z) \\ &\quad = -A + A\kappa z + Gz(1-\kappa z) + A(1-z) - D\kappa z(1-z) \\ &\quad = z\big(A(\kappa-1) + G(1-\kappa z) - \kappa D(1-z)\big) = zB, \end{aligned}$$

where

$$B = (\lambda+1)(\kappa-1) + (1-2\gamma-\lambda)(1-\kappa z) - \kappa(1-2\delta-\lambda)(1-z).$$

Multiply by $\delta+\lambda$ and use $\kappa(\delta+\lambda) = \gamma+\lambda$:

$$\begin{aligned}(\delta+\lambda)B = (\lambda+1)(\gamma-\delta) &+ (\delta+\lambda)(1-2\gamma-\lambda)(1-\kappa z)\\ &- (\gamma+\lambda)(1-2\delta-\lambda)(1-z).\end{aligned}$$

The right-hand side is affine in z. Its constant term is

$$\begin{aligned}K &= (\lambda+1)(\gamma-\delta) + (\delta+\lambda)(1-2\gamma-\lambda) - (\gamma+\lambda)(1-2\delta-\lambda)\\ &= \lambda\gamma - \lambda\delta + \gamma - \delta + \delta - 2\gamma\delta - \lambda\delta + \lambda - 2\gamma\lambda - \lambda^2\\ &\quad - \gamma + 2\gamma\delta + \gamma\lambda - \lambda + 2\delta\lambda + \lambda^2.\end{aligned}$$

The pairs $\lambda\gamma + \gamma\lambda - 2\gamma\lambda$, $-\lambda\delta - \lambda\delta + 2\delta\lambda$, $-2\gamma\delta + 2\gamma\delta$, $-\lambda^2 + \lambda^2$, $\lambda - \lambda$, $\gamma - \gamma$ and $-\delta + \delta$ each sum to zero, so $K = 0$.

The coefficient of z receives $-\kappa z$ from the second summand and $-(1-z)$ from the third:

$$\begin{aligned}&- (\delta+\lambda)(1-2\gamma-\lambda)\kappa + (\gamma+\lambda)(1-2\delta-\lambda)\\ &\quad = (\gamma+\lambda)\big(-(1-2\gamma-\lambda) + (1-2\delta-\lambda)\big)\\ &\quad = 2(\gamma+\lambda)(\gamma-\delta).\end{aligned}$$

Hence $(\delta+\lambda)B = 2z(\gamma+\lambda)(\gamma-\delta)$, and $zB = -2z(\gamma+\lambda)S$. Adding the three contributions yields $P(t) = S\,Q(t)$.

This is an AI-generated solution to a problem that was prompted by Paata Ivanisvili in Cursor with Grok 4.7.

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